INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 6: Conclusion and Future Work

by

TAEHONG KWON

Student number: 28576810

submitted in accordance with the requirements for the degree of

DOCTOR OF PHILOSOPHY

in the subject

COMPUTER SCIENCE

at the

UNIVERSITY OF SOUTH AFRICA

SUPERVISOR: Prof. Ernest Mnkandla

CO-SUPERVISOR: Prof. Donatien Koulla Moulla

CO-SUPERVISOR: Dr. David Sena Attipoe

October 2026

6 Conclusion and Future Work

Chapter 5 interpreted the findings and judged whether the study met its objectives. The tables of Chapter 4 form the record.

6.1 Answers to the research questions

The thesis proposed EndorseRank, PageRank with the edge weights of Adaptive Weighted PageRank (AWP) on a latest-allowance endorsement graph, and evaluated it beside AWP itself (T. Do et al., 2023), a coupled walk over both edge types and the raw degrees these walks smooth, on Arbitrum One contracts from October 2023 to June 2026, with holdout windows W0 (April to June 2026) and W1 (July to September 2026).

RQ1 asked whether EndorseRank and AWP rank the contracts of one sample differently. They do, and H1 is supported, largely by construction (Section 4.4). Ties play almost no part in the difference, but whether the two reflect different kinds of reputation rests mainly on the holdouts of RQ4.

RQ2 asked how closely each score agrees with allowance and transfer statistics in the same window. Each score follows the edges it walks on (Table 4.3), which supports H2 only narrowly, because the proxies come from the same events as the graphs.

RQ3 asked what each score costs to compute. The cost follows the edge count, so EndorseRank, on the sparsest graph, is the cheapest score, and the coupled walks cost more than AWP (Table 4.1); H3 is supported, largely by construction. On subsamples of the cohort the time ratio follows the edge ratio only at the two largest stages (Table 4.2).

RQ4 asked whether scores frozen at t_1 predict the new approval pairs and new transfer senders that spender contracts gain afterwards, and whether any PageRank predicts them better than the raw degree it smooths. The first part holds in both windows, and the edge type decides which label a score

anticipates (Tables 4.10 and 4.14). The second differs by walk, in contrasts computed after the labels were known: AWP ranks later new senders above the transfer in-degree in both windows, and EndorseRank ranks later approval pairs below the in-approve degree in both (Tables 4.11 and 4.16). EndorseRank matches its closed form at depth one, and deeper spenders show multi-hop propagation, which did not lift it above the count (Section 5.3.2); with AWP's activity restarts it ranked later approval pairs above the degree in both windows, a post hoc hypothesis, not a finding.

RQ5 asked whether a coupled PageRank over both edge types improves on a walk over the transfers alone, and whether it ranks contracts better than either layer walked alone. Both answers are no, on the revised scores as in the registered analysis (Table 8.12). Under rule A, C-PR meets the allowance condition of the primary part but fails its transfer condition and the secondary part (Section 4.3.1; Table 4.11). Under rule B, its exploratory gain over the transfer walk on new approval pairs in W0 recurred in W1, but its loss on new senders exceeded the margin, and on the traders' liquidation-free close rate a loss beyond it could not be ruled out, so the first answer is no; against AWP the loss on new senders was clear (Table 4.15). The seeded ablation matches the transfer walk on new senders: allowance edges contribute when the walk moves along them, and not as a seed.

In a comparison registered with rule B that adds no research question (Section 5.1.6), the approval count at the freeze led the transfer count on later liquidation avoidance, the outcome built from neither edge type that bears on counterparty risk, in W0, but in W1, the primary window, the interval of the lead includes zero ($P = +0.084$, 97.5% interval $[-0.026, 0.188]$; Section 4.6.3). The like-for-like PageRank pair is not resolved, and the approval count is lower on the profitable share of closes, so its signal concerns forced-liquidation avoidance, not trading success. Every trader is a contract account, so account type is no confound, but the samples are small.

Claims C1–C6 (Section 4.7) state these findings formally.

The main research question asked what ERC-20 allowance edges add to PageRank as a reputation score for contract accounts on Arbitrum One, compared with the transfer walk and with the raw degrees that both walks smooth. Against the transfer walk they add a second construct, delegated authority, which orders the contracts differently and ranks later approval pairs higher; against the

in-approve degree the allowance walk with uniform restarts added nothing out of window.

6.2 Original contribution to knowledge

In the terms of the literature reviewed in Chapter 2, the five contributions of Section 1.9 establish the following.

Before this work, graph reputation on public ledgers ranked wallets by their transfers (H.-D. Do & Do, 2023; T. Do et al., 2023), by how risk spreads through transaction networks (D. Lin et al., 2025), or by explicit ratings (Almasoud et al., 2020; Bellini et al., 2020), and the reputation tooling deployed in DeFi scored the holder of an address from credentials (Human Passport, n.d.). None of it used the ERC-20 allowance, the on-chain record of a deliberate delegation of spending power (Vogelsteller & Buterin, 2015), as the edge of a reputation ranking, or ranked contract accounts by the approvals they receive; approvals had been used to cluster the addresses of one entity (Victor, 2020) and studied as a security risk (Wang et al., 2022). None evaluated on-chain reputation scores on one cohort against the checks of each construct and against the raw degrees they smooth, with confidence intervals, difference tests and temporal holdouts under decision rules fixed in advance. This thesis supplies both; PageRank and AWP's weighting are taken from the literature (T. Do et al., 2023; Page et al., 1999), EndorseRank changes only the edge, and the coupled operator changes the walk.

The new knowledge has five parts. What an on-chain reputation score of a contract measures depends on the edges it uses, and this can be checked on one cohort with quantified uncertainty. The allowance edge yields a score that is cheaper to update, because its graph is sparser, differs from the transfer-based score, and ranks later endorsements of spenders in two windows, a label on which the scores built on transfers alone stay lower. On few traders, the comparison registered with rule B found no count-level lead of allowance over transfer on later avoidance of forced liquidation in the primary window and left the PageRank pair unresolved. Two further steps that the literature suggested did not consistently improve results on these data: propagation predicts the future growth of

the graph better than the raw degree it smooths on transfer edges but worse on allowance edges, and a weighted combination of the two edge types beats neither its better single layer on both spender labels nor, under rule B, the transfer-only walk. Finally, the allowance-side result has a partial analytic explanation: at depth one EndorseRank is, up to a constant, a linear function of the owner-normalized in-strength, which divides each owner's endorsement over all the spenders it approves; deeper spenders show multi-hop propagation, yet over all spenders EndorseRank did not beat the count (Section 5.3.2). The construct-validity framing (Cronbach & Meehl, 1955; Messick, 1995) that makes these claims testable is part of the contribution, as is the open pipeline whose published summaries let others check every table, since whether decentralized systems miscalculate credit scores is a matter of public interest (Packin & Lev-Aretz, 2024).

The scope is deliberately narrow. The thesis does not show that on-chain reputation can replace collateral (Gudgeon, Perez, et al., 2020; Schär, 2021), that the results apply beyond contracts approved on the five tokens on Arbitrum One in the window studied, or that any score measures creditworthiness, default risk or trading skill, and it does not claim that EndorseRank resists Sybils: under uniform teleportation a closed farm of attacker addresses can inflate it cheaply (Section 5.6).

6.3 Future work

Each suggestion below corresponds to a limitation identified in Chapter 5. Any new score or weighting it leads to should be fixed before it is scored, replicated on registered new data and checked out of window against degree baselines.

- A registered test of the revised analysis. Its verdicts are a re-analysis (Table 8.12). A registration committed before the logs of a later window, such as October to December 2026, are extracted would fix its US-dollar amounts, five tokens, value slopes and allowance cap with the cohort rule, rules A and B and the comparison on traders, which needs more contract traders than W0 and W1 held. The value-slope check did not cover the holdout labels (Table 4.7), so the slopes and the cap should not be tuned on the new window.

- Propagation on a deeper sample. On the ego networks studied, EndorseRank reduces to a reweighted approval count for every depth-one spender (Section 5.3.2); a sample extracted two or more hops out would test whether any propagated score adds out-of-window value over the local count, on arrival labels and on labels that are not counts of arrivals. The variant with AWP's activity restarts should be tested against the in-approve degree on new data, separately for depth-one and deeper spenders, since only deeper spenders can show whether propagation, and not the restarts alone, adds to the count.
- Validate the results in other contexts. Running the same analysis on other rollups, on Layer-1 networks and over later windows would show whether the construct checks and holdout results extend beyond Arbitrum One and the months studied and stay stable across market conditions; because DeFi is heterogeneous (Schär, 2021; Werner et al., 2022), this is the main threat to external validity. Scoring externally owned accounts lies outside this thesis.
- Adversarial evaluation. The Sybil cost model is analytical, and no attack was carried out on chain (Section 5.6). Restricting teleportation and dangling redistribution to a vetted seed set, as TrustRank does (Gyöngyi et al., 2004), would remove the uniform inflow that closed farms exploit and should be evaluated under the same attacks. Empirical red-teaming with wash approvals, coordinated spender farms and strategic revocation would measure the realized cost-effectiveness of attacks under live fee markets and detection heuristics, and would test whether the Sybil-adjusted proxies remain informative under active adversaries.
- Coupling beyond a fixed transition-level mixture. Only a fixed λ with uniform teleportation on US-dollar layers with capped allowances was tried. Node-dependent or learned weights, per-layer damping, mixtures of the two stationary vectors, and layers weighted as in EndorseRank and AWP remain untested; any such design should report construct-specific checks and keep the auditability that sets graph propagation apart from opaque learned scores (Kotb, 2023; Packin & Lev-Aretz, 2024).
- Composition with attestation tooling. Filtering the approving owners through a Passport-style

gate (Human Passport, n.d.), which answers whether an address belongs to a distinct person, before EndorseRank orders the contracts by the authority those owners delegate is a plausible deployment; it should be evaluated on labeled Sybil data, with the gate's own weaknesses (Bordeianu & Popescu, 2026; Siddarth et al., 2020) in view.

- Deployment studies. Integrating EndorseRank with risk oracles and governance tooling would test whether its runtime profile can sustain production refresh cadences. Controlled experiments on fee tiers or monitoring cadence would let evidence replace the adoption hypotheses of Section 5.4.4.

6.4 Closing reflection

In adversarial, pseudonymous markets on-chain reputation does not remove the need for collateral (Gudgeon, Perez, et al., 2020), but it can make behavioral structure readable, and which structure it reads depends on the edges used. A transfer records economic flow and an allowance records delegated authority. PageRank will rank either graph, yet one ranking does not serve both constructs, and a walk over both graphs gained little over the right single graph on one label and lost to it on the other. For a protocol or researcher who needs a fast, auditable signal of which contracts owners trust, the allowance edge is the one to use on data like those studied here. The approval count is its cheapest form and on these data ranked later approvals better than EndorseRank with uniform restarts; EndorseRank is the propagated form for those who want the endorsers to count as well, with its exposure to cheap inflation by closed farms in mind (Section 5.6). For describing transfer hubs and anticipating their inflow, the transfer edge is the one to use, and there the walk improved on the count. For screening counterparties against forced liquidation, the approval count led the transfer count among contract traders in W0 but not resolvably in the primary window; any such signal concerns liquidation avoidance, not trading success, and needs more traders to settle.